

Categories, Proofs, and Processes: Assignment 4

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Question 1

Proposition 1.1. *The covariant Hom-functor $\mathcal{C}(A, -) : \mathcal{C} \rightarrow \mathbf{Set}$ preserves all finite limits.*

Proof. Let $D : \mathcal{J} \rightarrow \mathcal{C}$ be a diagram where $\mathcal{J}$ is a finite category. Suppose the cone $(C, (c_1, c_2, \dots, c_N))$ to D is a finite limit. By being a cone, for any $f : I \rightarrow J$ in $\mathcal{J}$, we have

$$\begin{array}{ccc} D I & \xrightarrow{D f} & D J \\ & \swarrow c_I \quad \searrow c_J & \\ & C & \end{array}$$

$$D f \circ c_I = c_J$$

We show that $(\mathcal{C}(A, C), (\mathcal{C}(A, c_1), \dots, \mathcal{C}(A, c_N)))$ is a cone and a limit to $\mathcal{C}(A, -) \circ D : \mathcal{J} \rightarrow \mathbf{Set}$.

$$\mathcal{C} \xrightarrow{\mathcal{C}(A, -)} \mathbf{Set}$$

$$\begin{array}{ccc} C & \xrightarrow{\quad} & \mathcal{C}(A, C) \\ c_I \downarrow & \xrightarrow{\quad} & \downarrow \mathcal{C}(A, c_I) \\ D I & \xrightarrow{\quad} & \mathcal{C}(A, D I) \\ D f \downarrow & \xrightarrow{\quad} & \downarrow \mathcal{C}(A, D f) \\ D J & \xrightarrow{\quad} & \mathcal{C}(A, D J) \end{array}$$

Suppose we have an arbitrary $f : I \rightarrow J$ in $\mathcal{J}$, so we get

$$\mathcal{C}(A, (D f)) : \mathcal{C}(A, (D I)) \rightarrow \mathcal{C}(A, (D J))$$

Consider the morphisms $D I \xleftarrow{c_I} C \xrightarrow{c_J} D J$, so

$$\mathcal{C}(A, c_I) : \mathcal{C}(A, C) \rightarrow \mathcal{C}(A, (D I)) := h \mapsto c_I \circ h$$

$$\mathcal{C}(A, c_J) : \mathcal{C}(A, C) \rightarrow \mathcal{C}(A, (D J)) := h' \mapsto c_J \circ h'$$

$$\begin{array}{ccc}
\mathcal{C}(A, D I) & \xrightarrow{\mathcal{C}(A, D f)} & \mathcal{C}(A, D J) \\
& \nwarrow \mathcal{C}(A, c_I) \quad \nearrow \mathcal{C}(A, c_J) & \\
& \mathcal{C}(A, C) &
\end{array}$$

Then the diagram commutes since

$$\begin{aligned}
\mathcal{C}(A, (D f)) \circ \mathcal{C}(A, c_I) &= (h \mapsto (D f) \circ h) \circ (h' \mapsto c_I \circ h') \\
&= h \mapsto (D f) \circ c_I \circ h \\
&= h \mapsto c_J \circ h \\
&= \mathcal{C}(A, c_J)
\end{aligned} \tag{1}$$

So $(\mathcal{C}(A, C), (\mathcal{C}(A, c_1), \dots, \mathcal{C}(A, c_N)))$ is a cone to $\mathcal{C}(A, -) \circ D$.

To show the limit, consider $\mathcal{C}(A, C)$ in **Set** and let (B, γ) be some other cone to $\mathcal{C}(A, -) \circ D$, where γ is

$$\{\gamma_I : B \rightarrow \mathcal{C}(A, (D I))\}_{I \in \text{Ob}(\mathcal{J})}$$

Since $(C, (c_1, c_2, \dots, c_N))$ is a limit to D , for any cone (A, δ) to D , there is a unique arrow $a : A \rightarrow C$ such that $\forall I \in \text{Ob}(\mathcal{J}), c_I \circ a = \delta_I$.

Note that $\forall b \in B, \gamma_I(b) \in \mathcal{C}(A, (D I))$. Then for each $b \in B$, we have the cone

$$(A, \{\gamma_I(b) : A \rightarrow D I\}_{I \in \text{Ob}(\mathcal{J})})$$

From being the limit, there is a unique map $u_b : A \rightarrow C$ such that

$$\begin{array}{ccc}
& & D I \\
& \nearrow \gamma_I(b) & \uparrow c_I \\
A & \xrightarrow{u_b} & C
\end{array}$$

$$\forall I \in \text{Ob}(\mathcal{J}), c_I \circ u_b = \gamma_I(b)$$

By defining the arrow $g : B \rightarrow \mathcal{C}(A, C)$ such that $\forall b \in B, g(b) = u_b$, g is the unique arrow from the cone (B, γ) to $(\mathcal{C}(A, C), (\mathcal{C}(A, c_1), \dots, \mathcal{C}(A, c_N)))$, determined by the uniqueness of each u_b .

$$\begin{array}{ccc}
& & \mathcal{C}(A, D I) \\
& \nearrow \gamma_I & \uparrow \mathcal{C}(A, c_I) \\
B & \xrightarrow{g} & \mathcal{C}(A, C)
\end{array}$$

Thus, $(\mathcal{C}(A, C), (\mathcal{C}(A, c_1), \dots, \mathcal{C}(A, c_N)))$ is the finite limit to $\mathcal{C}(A, -) \circ D$.

□

Corollary 1.2. *Since all representable functors are naturally isomorphic to the covariant Hom-functor at A , $\mathcal{C}(A, -) : C \rightarrow \mathbf{Set}$ for some A , representable functors then preserve finite limits.*

Question 2

Proposition 2.3. *The natural transformation $\Delta : \text{Id} \rightarrow F$ given by arrows $\Delta_X : X \rightarrow X \times X := x \mapsto (x, x)$ is the only natural transformation of the type $\text{Id} \rightarrow F$.*

Proof. Δ is a natural transformation since the following naturality square commutes, where $f : A \rightarrow B$ is an arbitrary arrow in **Set**, and $f(a) = b \in B$, with the mappings as shown, for every $a \in A, A \in \text{Ob}(\mathbf{Set})$.

$$\begin{array}{ccccc}
 \text{Id} & \text{Id } A = A & \xrightarrow{\text{Id } f=f} & \text{Id } B = B & \\
 \downarrow \Delta & \downarrow \Delta_A & & & \downarrow \Delta_B \\
 F & F A = A \times A & \xrightarrow{F f=f \times f} & F B = B \times B & \\
 & & & & \\
 & & a \mapsto b & & \\
 & & \downarrow & & \downarrow \\
 & & (a, a) \mapsto (b, b) & &
 \end{array}$$

Suppose $\Gamma : \text{Id} \rightarrow F$ is another natural transformation. Then for any function $f : \{*\} \rightarrow A$ on the singleton set where $f(*) = a \in A$, the following diagram commutes.

$$\begin{array}{ccccc}
 \text{Id} & \text{Id } \{*\} = \{*\} & \xrightarrow{\text{Id } f=f} & \text{Id } A = A & \\
 \downarrow \Gamma & \downarrow \Gamma_{\{*\}} & & & \downarrow \Gamma_A \\
 F & F \{*\} := \{*\} \times \{*\} & \xrightarrow{F f=f \times f} & F A = A \times A & \\
 & & & & \\
 & & * \mapsto a & & \\
 & & \downarrow & & \downarrow \\
 & & (*, *) \mapsto (a, a) & &
 \end{array}$$

$= \{(*, *)\}$

In other words, we must have

$$\forall A \in \text{Ob}(\mathbf{Set}) : \Gamma_A(a) = (f \times f)(*, *) = (f(*), f(*)) = (a, a)$$

Then $\forall A, \Gamma_A = \Delta_A$, so Δ is the unique natural transformation. □

Question 3

Define

$$\begin{aligned} \text{eval} &: \mathcal{D}^{\mathcal{C}} \times \mathcal{C} \rightarrow \mathcal{D} \\ \text{eval} &:: (F : \mathcal{C} \rightarrow \mathcal{D}, C) \mapsto F \ C, \quad (\Delta : F \rightarrow G, h : C \rightarrow C') \mapsto (k : F \ C \rightarrow G \ C') \\ \mathcal{D}^{\mathcal{C}} \times \mathcal{C} &\xrightarrow{\text{eval}} \mathcal{D} \end{aligned}$$

$$\begin{array}{ccc} (F : \mathcal{C} \rightarrow \mathcal{D}, C) & \xrightarrow{\quad} & F \ C \\ (\Delta : F \rightarrow G, h : C \rightarrow C') \downarrow & \xrightarrow{\quad} & \downarrow k : F \ C \rightarrow G \ C' \\ (F : \mathcal{C} \rightarrow \mathcal{D}, C') & \xrightarrow{\quad} & G \ C' \end{array}$$

Lemma 3.4. *eval : $\mathcal{D}^{\mathcal{C}} \times \mathcal{C} \rightarrow \mathcal{D}$ is a functor.*

Proof. We use the lemma to show that **eval** is a functor. Fix an object $F : \mathcal{C} \rightarrow \mathcal{D}$ in $\mathcal{D}^{\mathcal{C}}$ so

$$\text{eval}(F, -) : \mathcal{C} \rightarrow \mathcal{D}$$

$$\text{eval}(F, -) :: C \mapsto \text{eval}(F, C), \quad f \mapsto F \ f$$

Clearly each object C in $\mathcal{C}$ has a mapping to $F \ C$ in $\mathcal{D}$. For arrow $f : A \rightarrow B, g : B \rightarrow C$ in $\mathcal{C}$,

$$\begin{aligned} \text{eval}(F, g \circ f) &= F(g \circ f) \\ &= F \ g \circ F \ f \quad (\text{by functoriality}) \\ &= \text{eval}(F, g) \circ \text{eval}(F, f) \end{aligned} \tag{2}$$

$$\text{eval}(F, 1_A) = F \ 1_A = 1_{F \ A} = 1_{\text{eval}(F, A)}$$

So **eval**($F, -$) satisfies functoriality.

Now fix an object C in $\mathcal{C}$ so

$$\text{eval}(-, C) : \mathcal{D}^{\mathcal{C}} \rightarrow \mathcal{D}$$

$$\text{eval}(-, C) :: F \mapsto \text{eval}(F, C), \quad t \mapsto t \ C$$

Each functor $F : \mathcal{C} \rightarrow \mathcal{D}$ in $\mathcal{D}^{\mathcal{C}}$ has a mapping to $F \ C$ in $\mathcal{D}$. For each natural transformation $\Delta : F \rightarrow G, \Gamma : G \rightarrow H$ in $\mathcal{D}^{\mathcal{C}}$,

$$\begin{aligned} & \begin{array}{ccccc} & & (\Gamma \circ \Delta)_C & & \\ & \nearrow \Delta_C & & \searrow \Gamma_C & \\ F \ C & \xrightarrow{\quad} & G \ C & \xrightarrow{\quad} & H \ C \end{array} \\ \text{eval}(\Gamma \circ \Delta, C) &= (\Gamma \circ \Delta)_C : F \ C \rightarrow H \ C \\ &= (\Gamma_C : G \ C \rightarrow H \ C) \circ (\Delta_C : F \ C \rightarrow G \ C) \\ &= \text{eval}(\Gamma, C) \circ \text{eval}(\Delta, C) \\ \text{eval}(1_F, C) &= 1_{F \ C} = 1_{\text{eval}(F, C)} \end{aligned} \tag{3}$$

So **eval**($-, C$) is a functor, and the first part of the lemma is satisfied.

Suppose we have a natural transformation $\Delta : F \rightarrow G$ in $\mathcal{D}^{\mathcal{C}}$, and an arrow $f : C \rightarrow C'$ in $\mathcal{C}$.

$$\begin{array}{ccc} \text{eval}(F, C) & \xrightarrow{\text{eval}(F, f)} & \text{eval}(F, C') \\ \text{eval}(\Delta, C) \downarrow & & \downarrow \text{eval}(\Delta, C') \\ \text{eval}(G, C) & \xrightarrow{\text{eval}(G, f)} & \text{eval}(G, C') \end{array}$$

Then the diagram commutes since the naturality square commutes with Δ being a natural transformation.

$$\begin{array}{ccc} F & C & \xrightarrow{f} F & C' \\ \Delta_C \downarrow & & & \downarrow \Delta_{C'} \\ G & C & \xrightarrow{f} G & C' \end{array}$$

$$\begin{aligned} \text{eval}(F, f) \circ \text{eval}(\Delta, C') &= (F \ f) \circ \Delta'_C \\ &= (G \ f) \circ \Delta_C \quad (\text{by naturality}) \\ &= \text{eval}(G, f) \circ \text{eval}(\Delta, C) \end{aligned} \tag{4}$$

The second part of the lemma is satisfied, and $\text{eval} : \mathcal{D}^{\mathcal{C}} \times \mathcal{C} \rightarrow \mathcal{D}$ is a functor. $\square$

Let $G : \mathcal{E} \times \mathcal{C} \rightarrow \mathcal{D}$ be an arbitrary functor. Define

$$\tilde{G} : \mathcal{E} \rightarrow \mathcal{D}^{\mathcal{C}}$$

$$\tilde{G} :: A \mapsto G(A, -), \quad (f : A \rightarrow B) \mapsto (\Delta : G(A, -) \rightarrow G(B, -))$$

where $G(A, -) : \mathcal{C} \rightarrow \mathcal{D}$ is a functor for each A in $\mathcal{E}$.

Lemma 3.5. *For a given $G : \mathcal{E} \times \mathcal{C} \rightarrow \mathcal{D}$, $\tilde{G} : \mathcal{E} \rightarrow \mathcal{D}^{\mathcal{C}}$ is the unique functor where the following diagram commutes.*

$$\begin{array}{ccc} \mathcal{D}^{\mathcal{C}} \times \mathcal{C} & \xrightarrow{\text{eval}} & \mathcal{D} \\ \tilde{G} \times 1_{\mathcal{C}} \uparrow & \nearrow G & \\ \mathcal{E} \times \mathcal{C} & & \end{array}$$

Proof. Any object A in $\mathcal{E}$ will get mapped to $G(A, -)$ in $\mathcal{D}^{\mathcal{C}}$. Suppose $A \xrightarrow{f} A' \xrightarrow{g} A''$ in $\mathcal{E}$. Then

$$\begin{aligned} \forall C \in \text{Ob}(\mathcal{C}), ((\tilde{G} \ g) \circ (\tilde{G} \ f))(C) &= (\tilde{G} \ g)(C) \circ (\tilde{G} \ f)(C) \\ &= (\Delta'_C : G(A', C) \rightarrow G(A'', C)) \circ (\Delta_C : G(A, C) \rightarrow G(A', C)) \\ &= \Gamma_C : G(A, C) \rightarrow G(A'', C) \\ &= \tilde{G}(g \circ f)(C) \end{aligned} \tag{5}$$

$$\tilde{G} \ 1_A = 1_{G(A, -)} = 1_{\tilde{G} \ A}$$

So $\tilde{G} : \mathcal{E} \rightarrow \mathcal{D}^{\mathcal{C}}$ is a functor.

Let (E, C) be an object in $\mathcal{E} \times \mathcal{C}$.

$$\text{eval} \circ (\tilde{G} \times 1_{\mathcal{C}})(E, C) = \text{eval}(G(E, -), C) = G(E, C)$$

Let (f, g) be an arrow in $\mathcal{E} \times \mathcal{C}$.

$$\begin{aligned} \mathbf{eval} \circ (\tilde{G} \times 1_{\mathcal{C}})(f : A \rightarrow A', g : C \rightarrow C') &= \mathbf{eval}(\Delta : G(A, -) \rightarrow G(A', -), g : C \rightarrow C') \\ &= k : G(A, C) \rightarrow G(A', C') \\ &= G(f : A \rightarrow A', g : C \rightarrow C') \end{aligned} \tag{6}$$

So $\mathbf{eval} \circ (\tilde{G} \times 1_{\mathcal{C}}) = G$.

To show uniqueness, suppose there is some other functor $H : \mathcal{E} \rightarrow \mathcal{D}^{\mathcal{C}}$ such that $\mathbf{eval} \circ (H \times 1_{\mathcal{C}}) = G$. Then for each object (E, C) in $\mathcal{E} \times \mathcal{C}$,

$$\mathbf{eval} \circ (H \times 1_{\mathcal{C}})(E, C) = \mathbf{eval}(H(E), C) = H(E)(C) = G(E, C)$$

So $\forall (E, C) \in \text{Ob}(\mathcal{E} \times \mathcal{C}), H(E)(C) = G(E, C)$. Then for each arrow (f, g) in $\mathcal{E} \times \mathcal{C}$,

$$\begin{aligned} \mathbf{eval} \circ (H \times 1_{\mathcal{C}})(f : A \rightarrow A', g : C \rightarrow C') &= \mathbf{eval}(H f : H A \rightarrow H A', g : C \rightarrow C') \\ &= k : H A C \rightarrow H A' C' \\ &= G(f : A \rightarrow A', g : C \rightarrow C') \end{aligned} \tag{7}$$

So $\forall (f, g) \in \text{Ob}(\mathcal{E} \times \mathcal{C}), H(f)(g) = G(f, g)$. Thus, $H = \tilde{G}$ is unique. $\square$

Proposition 3.6. *Func($\mathcal{C}, \mathcal{D}$) is an exponential object in $\mathbf{Cat}$.*

Proof. By Lemmas 3.4, 3.5, there is an object $\mathcal{D}^{\mathcal{C}}$ and morphism $\mathbf{eval} : \mathcal{D}^{\mathcal{C}} \times \mathcal{C} \rightarrow \mathcal{D}$, where for every $G : \mathcal{E} \times \mathcal{C} \rightarrow \mathcal{D}$, there is a unique morphism $\tilde{G} : \mathcal{E} \rightarrow \mathcal{D}^{\mathcal{C}}$ such that $\mathbf{eval} \circ (\tilde{G} \times 1_{\mathcal{C}}) = G$. Func($\mathcal{C}, \mathcal{D}$) is an exponential object in $\mathbf{Cat}$. $\square$

Question 4

Optional